

PROBLEM STATEMENTS



What are SDG goals

The United Nations Sustainable Development Goals (SDGs) are a universal call to action to end poverty, protect the planet, and ensure that all people enjoy peace and prosperity by 2030. Adopted by all United Nations Member States in 2015, these 17 interlinked goals serve as a blueprint for a more sustainable future for both the planet and its inhabitants. The SDGs address the global challenges we face, including those related to poverty, inequality, climate change, environmental degradation, peace, and justice. The goals are designed to be broad in scope yet specific in objectives, providing a comprehensive framework for governments, private sector, civil society, and individuals to contribute towards achieving them. The United Nations established these goals to mobilize global efforts around a common set of targets and indicators that promote human well-being, environmental sustainability, and economic prosperity, ensuring no one is left behind.

What you need to do?

As part of our initiative to integrate innovation with global development, each student is tasked with selecting one United Nations Sustainable Development Goal (SDG) to focus on. Your challenge is to deeply understand the intricacies of your chosen goal and craft a precise problem statement addressing a specific issue within that goal. Following this, you will use your technical prowess to design an Artificial Intelligence (AI) and Machine Learning (ML) solution aimed at resolving this issue. This project is an opportunity to apply AI and ML to real-world problems, promoting social impact while enhancing your skills. It's about harnessing technology for positive change and sustainable

development. Choose a goal that you are passionate about, and let's use innovation to pave the way for a better future. This exercise will sharpen both your technical abilities and your capacity to think critically about the application of technology in solving global challenges

The 17 SDG goals

No Poverty - End poverty in all its forms everywhere.

Zero Hunger - End hunger, achieve food security and improved nutrition, and promote sustainable agriculture.

Good Health and Well-being - Ensure healthy lives and promote well-being for all at all ages.

Quality Education - Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

Gender Equality - Achieve gender equality and empower all women and girls.

Clean Water and Sanitation - Ensure availability and sustainable management of water and sanitation for all.

Affordable and Clean Energy - Ensure access to affordable, reliable, sustainable, and modern energy for all.

Decent Work and Economic Growth - Promote sustained, inclusive, and sustainable economic growth, full and productive employment, and decent work for all.

Industry, Innovation, and Infrastructure - Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation.

Reduced Inequality - Reduce inequality within and among countries.

Sustainable Cities and Communities - Make cities and human settlements inclusive, safe, resilient, and sustainable.

Responsible Consumption and Production - Ensure sustainable consumption and production patterns.

Climate Action - Take urgent action to combat climate change and its impacts.

Life Below Water - Conserve and sustainably use the oceans, seas, and marine resources for sustainable development.

Life on Land - Protect, restore, and promote sustainable use of terrestrial ecosystems, manage forests sustainably, combat desertification, and halt and reverse land degradation and halt biodiversity loss.

Peace, Justice, and Strong Institutions - Promote peaceful and inclusive societies for sustainable development, provide access to justice for all, and build effective, accountable, and inclusive institutions at all levels.

Partnerships for the Goals - Strengthen the means of implementation and revitalize the Global Partnership for Sustainable Development.

Example problem statements

SDG 2: Zero Hunger

Problem Statement: An AI solution to predict crop yields and optimize farming practices based on weather patterns, soil health, and crop management techniques. The goal is to increase agricultural productivity and reduce hunger in rural communities.

SDG 3: Good Health and Well-being

Problem Statement: An ML model to analyze patient data and predict the likelihood of developing chronic diseases such as diabetes, heart disease, or cancer. The solution should also provide personalized health recommendations to improve patient outcomes and well-being.

SDG 4: Quality Education

Problem Statement: An AI-powered personalized learning platform that adapts to the individual needs of students, providing customized learning paths, resources, and assessments to improve educational outcomes, especially for underprivileged children.

SDG 7: Affordable and Clean Energy

Problem Statement: A chatbot to help households and businesses optimize their energy consumption. The chatbot should provide personalized recommendations for reducing energy use, suggest energy-efficient appliances, and offer information on renewable energy options.

SDG 6: Clean Water and Sanitation

Problem Statement: A chatbot to educate communities about water conservation techniques and best practices for maintaining clean water. The chatbot should answer questions about water-saving methods, provide tips for reducing water usage, and raise awareness about the importance of sanitation.

SDG 12: Responsible Consumption and Production

Problem Statement: A chatbot that assists users in making sustainable purchasing decisions. The chatbot should provide information on eco-friendly products, offer tips for reducing waste, and guide users on how to recycle and repurpose items effectively.

SDG 13: Climate Action

Problem Statement: An AI model to predict and mitigate the impact of climate change by analyzing historical climate data, current environmental conditions, and human activities. The solution should help policymakers and organizations develop effective climate action plans.

SDG 6: Clean Water and Sanitation

Problem Statement: A machine learning model to monitor water quality in real-time using sensor data from various sources. The solution should detect contaminants and provide early warnings to prevent waterborne diseases and ensure access to safe drinking water.